

PROIECT CCSM

BAZA ROTATIVA DE PREZENTARE CU MOTOR DC

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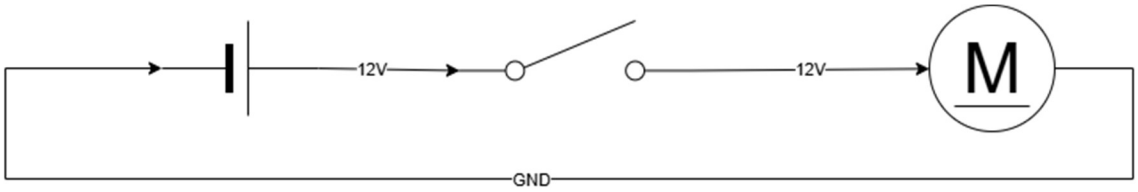
ETAPA 1

Denumirea sistemului:

BAZA ROTATIVA DE PREZENTARE CU MOTOR DC

Imagini/scheme ale sistemului:





Descrierea si enumerarea componentelor sistemului:

1. Motor DC Faulhaber 1331 012SR (Motor cu curent continuu 12V 4.11W 0.343A)
2. Sursa NA12P50 (Sursa alimentare Output: 12V 500mA)
3. Comutator Universal On/Off
4. Ansamblu Reductor 900:1

Descrierea sistemului, a modului de folosire si a utilitatii acestuia:

Sistemul este o baza rotativa de prezentare cu Ø200mm la 3-10 rotatii pe minut. Aceasta are scopul de a afisa obiecte de mici dimensiuni intr-o maniera estetica. Pentru a micsora rotatiile obiectului, aceasta foloseste un sistem de reductie prin roti dintate de plastic. Sistemul este tip de pinion-roata, in 4 trepte: 10:1 – 6:1 – 5:1 – 3:1.
Pentru a obtine aceste trepte de reductie, motorul este montat excentric in interiorul suportului, avand o raza de destindere a rotilor mai mare.
Pentru a fi folosita, se aseaza obiectul pe baza rotativa si se comuta butonul pe pozitia ON.

ETAPA 2

Enumerarea parametrilor tehnici ai motorului folosit:

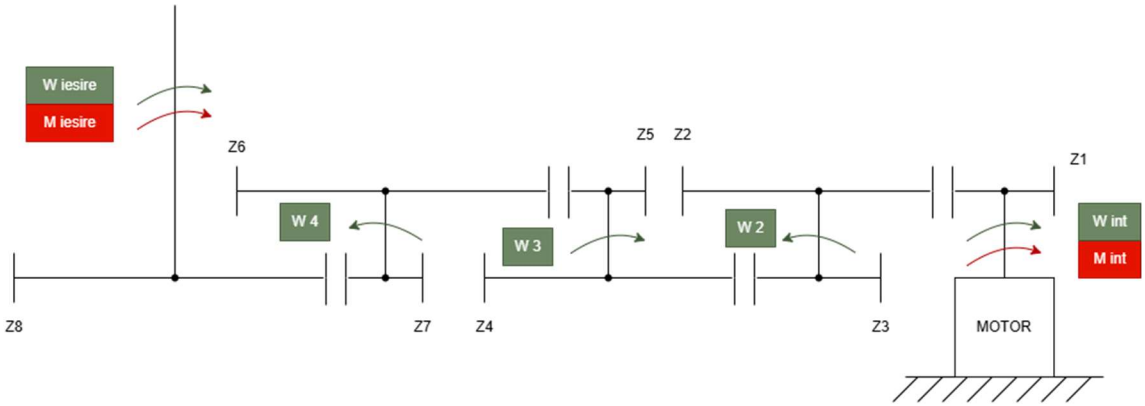
- tensiunea de alimentare: 12 V
- intensitatea curentului pentru mersul în gol: 0.0105 A
- turația pentru mersul în gol: 9910 rot/min
- momentul motor pentru mersul în gol: 9.94 N·mm
- diametrul motorului: 13 mm
- lungimea motorului: 31.2 mm
- diametrul arborelui motorului: 1.5 mm
- lungimea arborelui motorului: 6 mm

Foaia de catalog a motorului folosit:

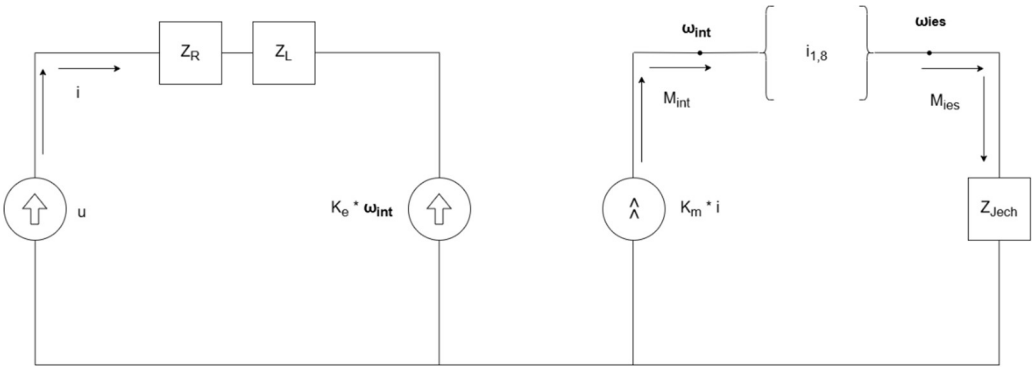
Aceasta poate fi gasita pe deplin la categoria ANEXE.

ETAPA 3

Schema de principiu:



Schema de impedante:



Ecuatiile circuitului:

$u=i*Z_R+i*Z_L+K_e*\omega_{int}$

$M_{int}=K_m*i$

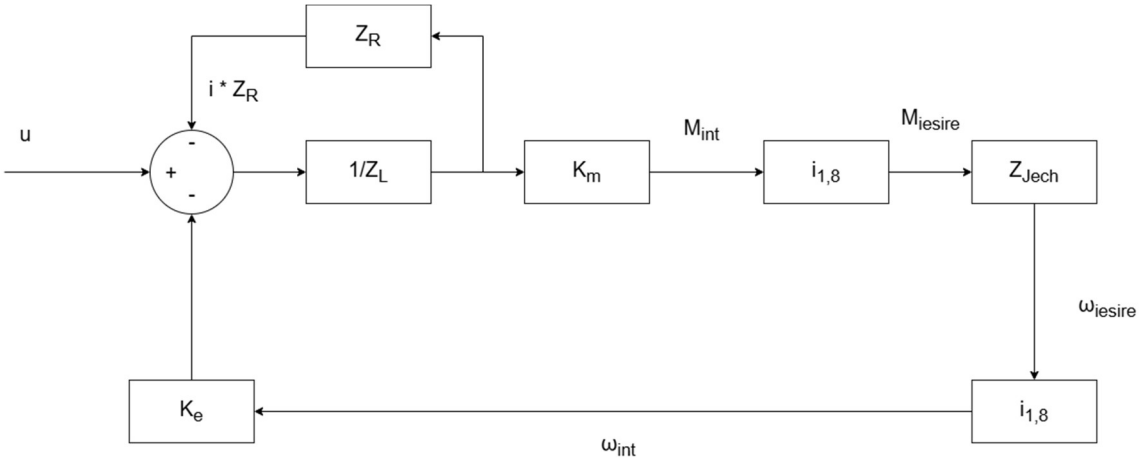
$i_{1,8}=\frac{\omega int}{\omega ies}$

$M_{ies}=M_{int}*i_{1,8}$

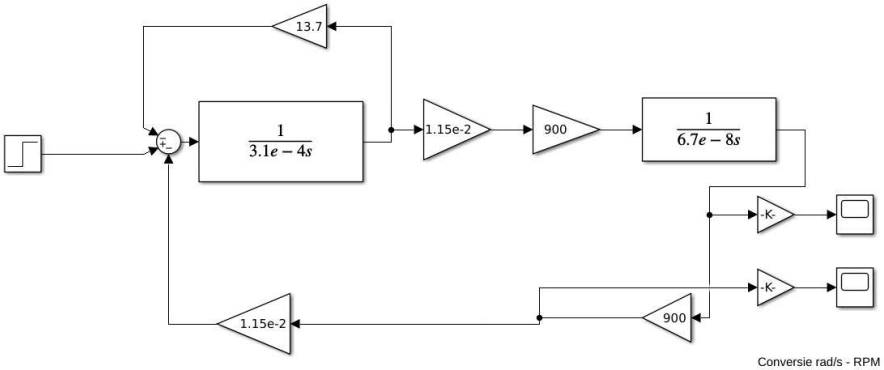
$\omega_{ies}=M_{ies}*Z_{Jech}$

ETAPA 4

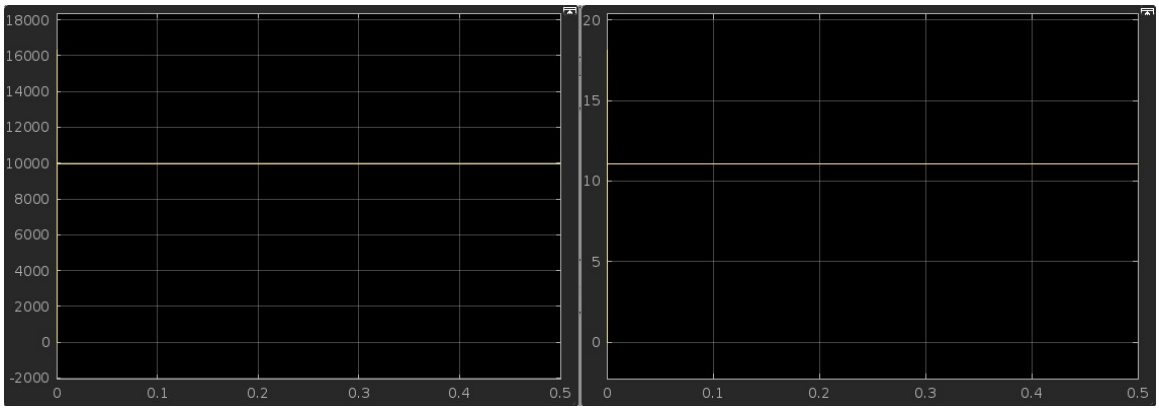
Schema de simulare:



Simularea (programul):



Grafice:



Interpretarea rezultatelor

Mecanismul are un comportament predictibil, in momentul in care este alimentat cu 12V, acesta ajunge fara o sarcina, considerand forta de frecare neglijabila, la aproximativ 10000RPM (viteza de intrare), foarte aproape de 9910RPM, cat este mentionat in specificatii, ajungand la 11RPM(viteza de iesire) la discul de 20cm, adica un raport de 909 :1, cauzat de neglijarea unor variabile/constante, cat si de marimea pasului de simulare si toleranta sa.
Putem considera simularea un succes in a oferi o varianta primitiva a unui mecanism mult mai complex.

ANEXE

DC-Micromotors

Precious Metal Commutation

3,44 mNm

5,14 W

Series 1331 ... SR

Values at 22°C and nominal voltage	1331 T		006 SR	012 SR	024 SR	
Nominal voltage	U_N		6	12	24	V
Terminal resistance	R		2,83	13,7	52,9	Ω
Rotor inductance	L		70	310	1 100	µH
Efficiency, max.	$\eta_{max.}$		80	79	79	%
No-load current, typ.	I_0		0,0226	0,0105	0,0055	A
No-load speed	n_0		10 700	9 910	10 500	min⁻¹
Stall torque	M_H		11,2	9,94	9,74	mNm
Rotor inertia	J		0,71	0,67	0,63	gcm²
Friction torque	M_R		0,12	0,12	0,12	mNm
Torque constant	k_M		5,33	11,5	21,7	mNm/A
Speed constant	k_n		1 790	833	439	min⁻¹/V
Slope of n-M curve	$\Delta n/\Delta M$		950	993	1 070	min⁻¹/mNm
Thermal resistance:						
– winding to housing	R_{th1}	11				K/W
– housing to ambient (external plastic flange)	$R_{th2\,p}$	28				K/W
– housing to ambient (external metal flange)	$R_{th2\,m}$	6				K/W
Thermal time constant:						
– winding	τ_{w1}	8,1				s
– housing (external plastic flange)	$\tau_{w2\,p}$	220				s
– housing (external metal flange)	$\tau_{w2\,m}$	48				s
Operating temperature range:						
– motor		-30 ... +85 (optional version -30 ... +125)				°C
– winding, max. permissible		+125				°C
Shaft bearings		sintered bearings	ball bearings, preloaded			
Shaft diameter		1,5	1,5			mm
Radial shaft load max.:						
– dynamic at 3 000 min⁻¹ (3 mm from bearing)		1,2	5			N
Axial shaft load max.:						
– dynamic at 3 000 min⁻¹		0,2	0,5			N
– static (shaft unsupported)		20	10			N
Shaft play, max.:						
– radial		0,03	0,015			mm
– axial		0,2	0			mm
Speed up to	$n_{max.}$	12 000				min⁻¹
Number of pole pairs		1				
Mass		19				g
Housing material		steel, nickel plated				
Magnet material		NdFeB				

Rated values for continuous operation						
Rated torque	M_N		1,96	3,44	3,31	mNm
Rated current (thermal limit)	I_N		0,4	0,343	0,174	A
Rated speed	n_N		8 750	5 260	5 650	min⁻¹

Note: Rated values are calculated with nominal voltage and at a 22°C ambient temperature. The $R_{th2\,p}$ value has been reduced by 0%.

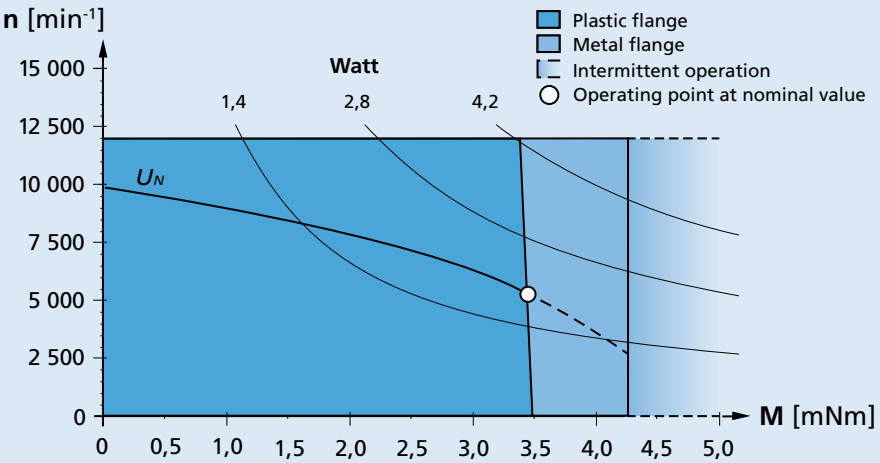
Note:

The diagram indicates the recommended speed in relation to the available torque at the output shaft for a given ambient temperature of 22°C.

The diagram shows the motor in different conditions of thermal coupling, i.e. mounted respectively on a plastic flange and a metal flange.

The nominal voltage (U_N) curve shows, up to the thermal limit, the operating point at nominal voltage for the motor mounted on a plastic flange. Higher torque can be achieved by further reducing the thermal resistance.

Any points of operation above the curve at nominal voltage will require a higher operating voltage. Any points below the nominal voltage curve will require less voltage.

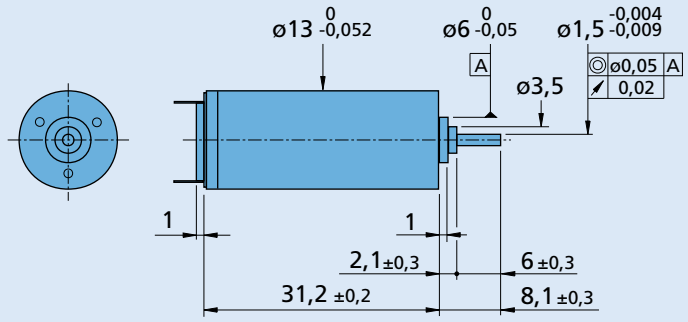


Recommended operation areas (example: nominal voltage 12V)

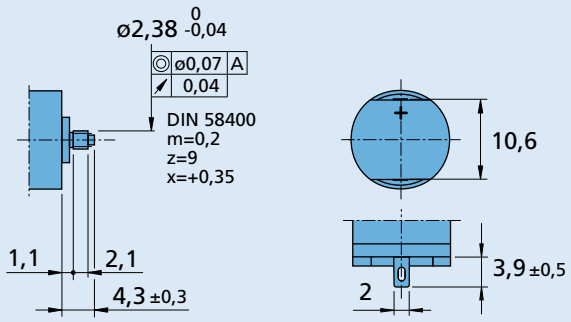
For notes on technical data and lifetime performance refer to "Technical Information".

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Dimensional drawing



1331 T ... SR



1331 E ... SR

Options

Example product designation: **1331T012SR-277**

Option	Type	Description
L	Twin Leads	For motors with twin leads (PVC), length 150 mm, red (+) / black (-)
4924	Twin Leads	For motors with twin leads (PVC), length 300 mm, red (+) / black (-)
X4924	Twin Leads	For motors with twin leads (PVC), length 600 mm, red (+) / black (-)
4925	Twin Leads	For motors with twin leads (PVC), length 150 mm, red (+) / black (-), with connector AMP 179228-2
X4925	Twin Leads	For motors with twin leads (PVC), length 300 mm, red (+) / black (-), with connector AMP 179228-2
Y4925	Twin Leads	For motors with twin leads (PVC), length 600 mm, red (+) / black (-), with connector AMP 179228-2
F	Single Leads	For motors with single leads (PTFE), length 150 mm, red (+) / black (-)
277	Bearings	2 preloaded ball bearings

Product combination

Precision Gearheads / Lead Screws	Encoders	Drive Electronics	Cables / Accessories
13A 14GPT 14/1 15/5 15/5 S	IE2-400	SC 1801 P SC 1801 S MC 3001 B MC 3001 P MC 3602 B MC 3603 S MC 5004 P	To view our large range of accessory parts, please refer to the “Accessories” chapter.

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